AI-Based Hand Gesture Recognition for Sign Language Communication

A Mini Project Report Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

By

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DECLARATION

The mini-project report entitled "**AI-Based Hand Gesture Recognition for Sign Language Communication**" is a record of bona fide work carried out by K.SAI SREE CHOWDARY-2420030168 in partial fulfillment of the requirements for the award of the B.Tech degree at K L (Deemed to be) University. The work presented in this report is original to the best of our knowledge, and all datasets, tools, code, and external sources used have been appropriately acknowledged.

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CERTIFICATE

This is to certify that the mini-project report entitled "**AI-Based Hand Gesture Recognition for Sign Language Communication**" is a bona fide work carried out by K.SAI SREE CHOWDARY-2420030168 under the supervision of DR.K.SWANTHANA during the academic year [2024–2028], in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

Signature of the HOD **Signature of the External Examiner**

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K.SAI SREE CHOWDARY-2420030168

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1. ABSTRACT

Sign language communication remains a challenge for deaf and hard-of-hearing individuals, particularly when people around them do not understand sign language. This project, “AI-Based Hand Gesture Recognition for Sign Language Communication,” develops a deep learning-based system for recognizing isolated Indian Sign Language (ISL) signs from video and converting the recognized signs into text and speech. The system uses the publicly available INCLUDE dataset, with 262 sign classes and an official partition of 3,121 training videos, 347 validation videos, and 816 test videos. Video preprocessing is performed using OpenCV, with each video uniformly sampled into 64 frames. MediaPipe extracts hand, body-pose, and selected facial landmarks, from which 469-dimensional geometric features and a four-dimensional availability mask are derived. The extracted features are processed using part-aware temporal one-dimensional convolutional neural network (Conv1D) encoders, availability-aware gated fusion, and four Head-Token Mixing Attention (HTMA) blocks to capture spatial and temporal relationships. The final classifier predicts one of 262 sign classes. Unlike conventional text analytics systems, this project uses structured visual and motion features rather than text representations such as TF-IDF or word embeddings. Evaluation on the locked test set achieved 98.04% Top-1 accuracy, 99.51% Top-5 accuracy, 98.09% Macro-F1, and 98.19% balanced accuracy. A FastAPI backend and React frontend present recognized signs as text and speech. The system demonstrates the potential of landmark-based deep learning to support accessible ISL communication, while continuous sign recognition and measured live-webcam performance remain future work.

The proposed system integrates temporal feature extraction and attention-based learning to capture the variations in hand movements and body posture across video frames. By combining part-aware temporal convolutional networks with availability-aware gated fusion and Head-Token Mixing Attention (HTMA) blocks, the model learns meaningful spatial and temporal patterns while handling variations in landmark visibility. The trained classifier predicts the corresponding sign from 262 classes and presents the recognized output as text, with text-to-speech functionality supporting audible communication. A FastAPI backend connects the recognition pipeline to a React-based user

interface, enabling users to interact with the system and view predictions. Experimental evaluation on the designated test set demonstrates a top-1 accuracy of 98.04%, a top-5 accuracy of 99.51%, a macro-F1 score of 98.09%, and a balanced accuracy of 98.19%. These results indicate the potential of landmark-based temporal modeling and attention mechanisms for accurate sign language recognition. The proposed approach provides a foundation for more accessible communication systems, with future enhancements focusing on continuous sign recognition, real-time webcam evaluation, and broader testing across different users and environments.

Team Name:

AI-Based Hand Gesture Recognition for Sign Language Communication

Team Members and Responsibilities:

1-K.SAI SREE CHOWDARY-TEAM LEAD

DATASET URL:https://zenodo.org/records/4010759

GITHUBURL:https://github.com/saisree-kosaraju/KLH-CSE-2026-27-2420030168-SignGestureAI

2. INTRODUCTION

Sign language is an important means of communication for deaf and hard-of-hearing individuals. However, communication barriers arise when people who use sign language interact with individuals who do not understand it. In educational institutions, hospitals, workplaces, and public service environments, the absence of a common communication method can make everyday interactions difficult. Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision provide opportunities to address this challenge by automatically recognizing sign language gestures from video and converting them into understandable text and speech.

The project, **"AI-Based Hand Gesture Recognition for Sign Language Communication,"** focuses on isolated Indian Sign Language (ISL) recognition using a temporal deep learning approach. Manually interpreting sign language videos is difficult to perform consistently at scale because signs involve variations in hand movements, body posture, facial features, signer characteristics, lighting, and background conditions. Automated recognition can process visual sequences systematically and learn meaningful spatial and temporal patterns from training data. This reduces dependence on manual interpretation and provides a foundation for accessible communication support.

2.1 Application Area and Problem Statement

The application area of this project is AI-based sign language recognition and assistive communication. The system is designed to recognize individual Indian Sign Language signs from video sequences and present the predicted sign as text and synthesized speech. The project uses the INCLUDE dataset and focuses on a vocabulary of 262 sign classes. Unlike text classification systems that process documents or sentences, the proposed system processes video frames and extracts structured landmark and motion-related features.

The main problem addressed is the difficulty of translating visual sign gestures into an understandable output for people unfamiliar with sign language. The system aims to provide a camera-based recognition approach without requiring gloves or specialized sensors. Its current

experimental scope is isolated-sign classification; recognizing continuous signing and translating complete sentences remain potential future extensions.

2.2 Target Users and Intended Use

The intended users include deaf and hard-of-hearing individuals, people communicating with sign language users, educators, students, and developers working on accessibility technologies. The application can support communication demonstrations, educational activities, and research into automated ISL recognition. It is intended as an assistive tool and should not be considered a complete replacement for qualified sign language interpreters.

2.3 Input and Predicted Output

The input to the system is a video containing an isolated Indian Sign Language gesture. OpenCV reads the video and uniformly samples 64 frames. MediaPipe extracts hand, body-pose, and selected facial landmarks from these frames. The pipeline derives 469-dimensional geometric features and a four-dimensional availability mask to represent gesture geometry and indicate the availability of detected body parts.

The model predicts one class from the 262 supported sign categories. The output is displayed as a text label, along with confidence and ranked Top-5 predictions through the application interface. The recognized label can also be converted into speech using the browser's Web Speech API.

2.4 Proposed Methodology

The proposed system follows a structured machine learning pipeline:

1. **Data collection and preparation:** The INCLUDE dataset supplies labeled Indian Sign Language videos. The official data partition used in the project contains 3,121 training videos, 347 validation videos, and 816 test videos across 262 classes.
2. **Video preprocessing:** OpenCV reads the videos, samples 64 frames uniformly, and prepares the frames for landmark extraction.

3. **Feature extraction and representation:** MediaPipe detects hand, pose, and selected face landmarks. Geometric features and an availability mask represent body-part information and missing detections.
4. **Model development and training:** Part-aware temporal Conv1D encoders learn motion patterns from different feature streams. Availability-aware gated fusion combines these representations, and four Head-Token Mixing Attention (HTMA) blocks learn temporal relationships across the frame sequence. A classifier generates predictions for the supported sign classes.
5. **Model evaluation:** The trained model is evaluated using Top-1 accuracy, Top-5 accuracy, macro-F1 score, and balanced accuracy on the locked test set.
6. **Application development:** A FastAPI backend serves model predictions, while a React-based frontend supports video upload or camera recording and displays the results. Speech output is provided through the browser's Web Speech API.

The evaluated system achieved 98.04% Top-1 accuracy, 99.51% Top-5 accuracy, 98.09% macro-F1 score, and 98.19% balanced accuracy on the 816-video test set. These metrics describe the evaluated test-set performance and do not represent measured live-webcam accuracy.

3. LITERATURE SURVEY

The literature survey for the project "AI-Based Hand Gesture Recognition for Sign Language Communication" examines existing research on isolated sign language recognition, landmark-based gesture classification, temporal deep learning, and continuous sign language translation. These studies provide the foundation for selecting the INCLUDE dataset and developing a model that combines structured body landmarks, temporal feature extraction, attention mechanisms, and text-to-speech output.

3.1 Review of Research Papers

1. INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition (2020)

Sridhar et al. introduced INCLUDE, a large-scale dataset designed to support isolated Indian Sign Language recognition, addressing the shortage of publicly available ISL benchmarks. The dataset contains approximately 4,287 videos covering 263 sign classes, with an additional 50-class subset called INCLUDE-50. The study evaluated deep learning approaches using visual features, including a frozen MobileNetV2 feature extractor with a bidirectional LSTM-based temporal model. The reported accuracy was 85.6% on the full dataset and 94.5% on INCLUDE-50.

The study provides a relevant benchmark for the present project, although its original experiments and data partition differ from the 262-class evaluation used in this implementation.

2. Word-Level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison (2020)

Li et al. introduced the WLASL dataset for word-level American Sign Language recognition, containing more than 2,000 sign categories performed by over 100 signers. The study compared visual appearance-based methods with pose-based approaches using video frames and body keypoints. It also proposed Pose-TGCN, which models spatial relationships between body joints and their temporal movement using graph convolutions. The experiments demonstrated the difficulty of maintaining recognition performance as the vocabulary increases, particularly for large-scale

recognition.

However, WLASL focuses on American Sign Language rather than Indian Sign Language, and its benchmark results cannot be directly compared with those obtained on INCLUDE.

3. Sign Pose-Based Transformer for Word-Level Sign Language Recognition (2022)

Boháček and Hrúz proposed SPOTER, a Transformer-based approach designed to recognize individual signs using estimated body and hand pose landmarks. The method uses pose normalization, a separate coordinate system for hand poses, and data augmentation to improve recognition while limiting computational requirements. It was evaluated on WLASL100, WLASL300, and LSA64, reporting top-1 recognition rates of 63.18%, 43.78%, and 100%,respectively.

The results demonstrate the value of temporal attention and pose normalization, but the experiments use different sign languages and datasets from the present project. The method also relies primarily on pose representations rather than the proposed project's explicit multi-stream geometry and availability-aware fusion.

4. SignBERT: Pre-Training of Hand-Model-Aware Representation for Sign Language Recognition (2021)

Hu et al. introduced SignBERT, a self-supervised pre-training framework that treats hand-pose sequences as visual tokens and incorporates hand structure, gesture state, temporal information, and hand chirality. The method learns useful representations by masking and reconstructing hand-pose tokens before fine-tuning the model for sign classification. It was evaluated on four public benchmarks: NMFs-CSL, SLR500, MSASL, and WLASL. On WLASL2000, the results reported in the project literature review include 39.40% top-1 accuracy using hands alone and 54.69% using combined hand and RGB predictions.

Although the approach demonstrates the benefits of hand-aware learning and pre-training, it requires an additional pre-training stage and does not specifically evaluate Indian Sign Language.

5. Sign Language Transformers: Joint End-to-End Sign Language Recognition and

Translation (2020)

Camgoz et al. proposed a Transformer-based architecture that jointly learns continuous sign language recognition and translation. The model combines a recognition component with a translation component using Connectionist Temporal Classification (CTC), allowing the system to learn from continuous sign sequences without requiring ground-truth timing for individual signs. The researchers evaluated their approach on the RWTH-PHOENIX-Weather-2014T dataset, which contains German Sign Language weather broadcasts. Their reported results include a test word error rate of 24.49 and a BLEU-4 score of 21.80 for the relevant recognition and translation systems.

This work provides a foundation for sequence-based sign recognition, but its focus is continuous German Sign Language and sentence translation rather than isolated Indian Sign Language classification.

6. Automatic Indian Sign Language Recognition Using MediaPipe Holistic and LSTM Network (2024)

Khartheesvar et al. investigated isolated Indian Sign Language word recognition using MediaPipe Holistic for landmark extraction and a Long Short-Term Memory (LSTM) network for temporal sequence modelling. The study evaluated the approach on INCLUDE and INCLUDE-50, making it particularly relevant to the present project. It reported 87.4% accuracy and an 86.6% macro-F1 score on INCLUDE, while the smaller INCLUDE-50 subset achieved 94.8% accuracy and 93.5% macro-F1.

The results demonstrate the usefulness of landmark-based temporal learning for ISL. However, the model uses an LSTM-based architecture rather than the present project's combination of part-aware temporal convolutional encoders, availability-aware gated fusion, geometric features, and Head-Token Mixing Attention (HTMA).

3.2 Comparative Analysis

The reviewed studies demonstrate several approaches to sign language recognition, each with different strengths and limitations.

Research work	Main approach	Dataset / language	Main limitation relevant to this project
INCLUDE (2020)	Visual features and CNN–BiLSTM modelling	INCLUDE / ISL	Original architecture does not use the proposed multi-stream fusion and HTMA design.
Li et al. (2020)	RGB models and pose-based temporal graph convolution	WLASL / ASL	Different sign language; performance varies with vocabulary size.
SPOTER (2022)	Pose-based Transformer	WLASL, LSA64	Primarily pose-driven; not evaluated on INCLUDE.
SignBERT (2021)	Hand-aware self-supervised Transformer	WLASL and other benchmarks	Requires pre-training and does not specifically target ISL.
Camgoz et al. (2020)	Joint Transformer recognition and translation	PHOENIX14T / German Sign Language	Addresses continuous signing and translation rather than isolated ISL classification.

Khartheesvar et al. (2024)	MediaPipe Holistic and LSTM		INCLUDE / ISL	Uses recurrent temporal modelling rather than the proposed gated multi-stream architecture.

The studies support the use of video-based features, pose landmarks, temporal modelling, and attention mechanisms for sign recognition. However, their datasets, sign languages, vocabulary sizes, and evaluation protocols differ. Consequently, their reported performance figures should not be treated as directly comparable rankings.

3.3 Research Gap and Motivation

The literature review identifies the following gaps that motivate the proposed project:

1. Limited research on Indian Sign Language: Many established methods are evaluated on American or German Sign Language, while fewer studies focus on large-vocabulary ISL recognition.
2. Limited integration of complementary features: Some systems use RGB appearance or a single concatenated pose representation. The proposed system separately encodes hand, pose, face, and geometric information.
3. Handling missing landmarks: In practical videos, some body parts may not be detected consistently. The proposed model uses an availability mask and gated fusion to account for feature availability.
4. Temporal modelling: CNN–BiLSTM, graph convolution, and Transformer architectures have demonstrated the importance of learning motion patterns. This project explores part-aware temporal Conv1D encoders followed by four HTMA blocks to model relationships across video frames.
5. Application-level communication support: Recognition systems need an accessible way to present their predictions. The proposed implementation integrates a FastAPI backend with a React interface and browser-based text-to-speech.

Based on these gaps, the project develops a landmark-based temporal deep learning pipeline using the INCLUDE dataset, with 262 sign classes. The evaluated system achieved 98.04% Top-1 accuracy, 99.51% Top-5 accuracy, 98.09% macro-F1, and 98.19% balanced accuracy on its locked test set of 816 videos. These results describe the project's own evaluation protocol and should not be interpreted as a controlled, direct comparison against the published results above.

3.4 References

[1] INCLUDE dataset and baseline study

Sridhar, A., Ganesan, R. G., Kumar, P., and Khapra, M. (2020). "INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition." Proceedings of the 28th ACM International Conference on Multimedia, pp. 1366–1375.

- DOI: https://doi.org/10.1145/3394171.3413528
- Dataset and baseline repository: https://github.com/AI4Bharat/INCLUDE

[2] WLASL dataset and recognition methods

Li, D., Rodriguez, C., Yu, X., and Li, H. (2020). "Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 1459–1469.

- Paper: https://openaccess.thecvf.com/content_WACV_2020/html/Li_Word-level_Deep_Sign_Language_Recognition_from_Video_A_New_Large-scale_WACV_2020_paper.html
- Dataset: https://dxli94.github.io/WLASL/

[3] SPOTER

Boháček, M., and Hrúz, M. (2022). "Sign Pose-Based Transformer for Word-Level Sign Language Recognition." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops, pp. 182–191.

- Paper:https://openaccess.thecvf.com/content/WACV2022W/HADCV/html/Bohacek_Sign_P

ose-Based_Transformer_for_Word-Level_Sign_Language_Recognition_WACVW_2022_paper.html

[4] SignBERT

Hu, H., Zhao, W., Zhou, W., Wang, Y., and Li, H. (2021). "SignBERT: Pre-Training of Hand-Model-Aware Representation for Sign Language Recognition." Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 11087–11096.

- Paper: https://openaccess.thecvf.com/content/ICCV2021/html/Hu_SignBERT_Pre-Training_of_Hand-Model-Aware_Representation_for_Sign_Language_Recognition_ICCV_2021_paper.html
- DOI: https://doi.org/10.1109/ICCV48922.2021.01090

[5] Sign Language Transformers

Camgoz, N. C., Koller, O., Hadfield, S., and Bowden, R. (2020). "Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 10023–10033.

- Paper: https://openaccess.thecvf.com/content_CVPR_2020/html/Camgoz_Sign_Language_Transformers_Joint_End-to-End_Sign_Language_Recognition_and_Translation_CVPR_2020_paper.html
- DOI: https://doi.org/10.1109/CVPR42600.2020.01004

[6] MediaPipe Holistic and LSTM for ISL

Khartheesvar, G., Kumar, M., Yadav, A. K., and Yadav, D. (2024). "Automatic Indian Sign Language Recognition Using MediaPipe Holistic and LSTM Network." Multimedia Tools and Applications, 83, pp. 58329–58348.

- DOI: https://doi.org/10.1007/s11042-023-17361-y
- Publication information: https://www.researchgate.net/publication/376719501_Automatic_Indian_sign_language_reco

gnition_using_MediaPipe_holistic_and_LSTM_network

4. PROBLEM STATEMENT AND OBJECTIVES

4.1 Problem Statement

To develop a machine-learning-based Indian Sign Language recognition system that classifies isolated sign gestures from video into 262 sign classes and converts the predicted sign labels into text and speech to support accessible communication.

4.2 Problem Description

Communication between Indian Sign Language users and people unfamiliar with sign language can be challenging in educational institutions, workplaces, hospitals, and public places. Manual interpretation of video gestures is difficult to scale consistently because signs involve temporal hand movements, body posture, and variations in recording conditions. This project uses landmark-based temporal deep learning to recognize isolated signs and provide text and speech output through a web application. The current implementation focuses on 262 isolated sign classes and does not yet support continuous sign-language sentence recognition.

4.3 Objectives

- Obtain and inspect the labelled INCLUDE Indian Sign Language dataset, documenting its source, video counts, class labels, and training, validation, and test partitions.
- Preprocess sign-language videos by uniformly sampling 64 frames and extracting hand, body-pose, and selected facial landmarks using MediaPipe.
- Generate numerical representations using geometric features and an availability mask to represent landmark information and missing detections.
- Develop and train a temporal deep learning model using part-aware Conv1D encoders, availability-aware gated fusion, and Head-Token Mixing Attention (HTMA) blocks.
- Evaluate the trained model on the locked test set using Top-1 accuracy, Top-5 accuracy, macro-F1 score, balanced accuracy, and a confusion matrix where applicable.
- Analyse recognition errors and discuss limitations related to signer variation, lighting, background conditions, missing landmarks, and generalization to unseen recording environments.

- Implement a FastAPI backend and React-based interface to accept video input, display predicted sign labels and confidence information, and provide text-to-speech output.
- Document the model's performance and identify future improvements, including continuous sign recognition and live-webcam evaluation.

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 Hardware Requirements

- Computer/Laptop: A modern multi-core CPU capable of processing video data and running deep learning models.
- RAM: Minimum 8 GB recommended; 16 GB or more is preferable for smoother dataset processing and model training.
- GPU: An NVIDIA GPU with CUDA support is recommended for faster PyTorch model training, but it is not mandatory.
- Storage: Sufficient disk space for the INCLUDE dataset, extracted landmarks, model checkpoints, source code, and evaluation results.
- Internet Connection: Required for downloading the dataset, Python packages, and required dependencies.
- Camera: A webcam is optional for recording sign videos through the demonstration interface.

5.2 Software Requirements

- Operating System: Windows, Linux, or macOS.
- Programming Language: Python 3.x.
- Development Environment: Visual Studio Code, Jupyter Notebook, or Google Colab.
- Deep Learning Framework: PyTorch for model development and training.
- Video Processing: OpenCV for reading and sampling video frames.
- Landmark Extraction: MediaPipe for detecting hand landmarks, body pose, and selected face landmarks.
- Numerical and Data Processing: NumPy and Pandas.
- Visualization and Evaluation: Matplotlib, scikit-learn, and other required evaluation libraries.
- Backend Framework: FastAPI for handling video uploads and returning prediction results.
- Frontend Framework: React with Vite for the user interface.
- Speech Output: Browser Web Speech API (speechSynthesis) for speaking the predicted sign label.
- Dataset: INCLUDE (Indian Sign Language dataset), using the project's selected 262 sign

classes and the official training, validation, and test partitions.

- Version Control: Git and GitHub for source-code management and project collaboration.

6. IMPLEMENTATION

6.1 Data and Task

The project, "AI-Based Hand Gesture Recognition for Sign Language Communication," focuses on isolated Indian Sign Language (ISL) recognition from video. The system classifies an input sign video into one of 262 sign classes and displays the predicted label as text, with an optional speech output.

The project uses the INCLUDE (Indian Sign Language) dataset, which contains videos representing Indian Sign Language signs. The implemented version uses 262 classes, excluding the secondnumber class from the original 263-class label set.

Dataset detail	**Description**
Dataset source	INCLUDE — AI4Bharat
Task	Video-based sign classification
Number of classes	262
Training videos	3,121
Validation videos	347
Test videos	816

Total videos	4,284
Input	Sign-language video
Output	Predicted sign label, confidence score, and Top-5 predictions

The dataset is divided into training, validation, and test sets. The training set is used to learn model parameters, the validation set is used for model selection, and the locked test set is used for final performance evaluation. The dataset's licence and usage conditions should be checked against the official dataset distribution before redistribution or publication.

6.2 Data Preprocessing and Feature Extraction

Since the project processes video and body landmarks rather than text, conventional NLP preprocessing methods such as tokenization, stop-word removal, stemming, lemmatization, and TF-IDF are not applicable.

The following preprocessing steps are implemented:

1. Video loading: OpenCV is used to read the input videos.
2. Frame sampling: Each video is represented by uniformly sampling 64 frames. Short videos are padded by repeating the last frame.
3. Landmark extraction: MediaPipe detects hand landmarks, body-pose landmarks, and selected face landmarks.
4. Feature generation: Each frame is represented using 534-dimensional landmark features, 469-dimensional geometry features, and a 4-dimensional availability mask.
5. Missing landmark handling: Availability information is used to indicate which landmark parts are detected. Pose-guided hand rescue is applied where possible.
6. Temporal representation: The extracted features preserve the sequence of movements across

the sampled frames, enabling the model to learn temporal patterns associated with different signs.

This preprocessing pipeline converts raw video into structured numerical features suitable for deep learning

6.3 Model Training and Evaluation

The project uses a custom temporal deep learning architecture implemented in PyTorch. It combines part-aware temporal Conv1D encoders, availability-aware gated fusion, and Head-Token Mixing Attention (HTMA) blocks.

Model training configuration

Parameter	Configuration
Framework	PyTorch
Optimizer	AdamW
Learning rate	2.5×10^{-4}
Weight decay	2×10^{-4}
Batch size	32
Maximum epochs	70

Learning-rate schedule	5 warm-up epochs followed by cosine decay
Minimum learning rate	10^{-6}
Gradient clipping	1.0
EMA decay	0.995
Loss function	Weighted cross-entropy with label smoothing of 0.10
Random seed	42
Early stopping patience	15 epochs

The model uses separate encoders for hand, pose, face, and geometry features. Five learned gates, guided by the availability mask, fuse the extracted representations into a 320-dimensional representation for each frame. Four HTMA blocks model temporal relationships. The resulting representation is passed to a classifier that predicts one of the 262 classes.

The EMA checkpoint selected at epoch 38 was chosen using validation performance. The locked test set was not used for training or model selection.

Evaluation metrics and results

The model is evaluated using Top-1 accuracy, Top-5 accuracy, macro-F1 score, and balanced

accuracy.

Metric	Test result
Top-1 accuracy	98.04%
Top-5 accuracy	99.51%
Macro-F1 score	98.09%
Balanced accuracy	98.19%
Correct predictions	800 out of 816

These results represent evaluation on the held-out test videos. They should not be interpreted as measured live-webcam accuracy. Continuous sign recognition and sentence-level translation are outside the current implementation.

6.4 Project workflow and modules

1. Data loading and exploratory analysis: The project uses the INCLUDE dataset and its official training, validation, and test partitions. Dataset metadata and label mappings are handled as part of the data-processing workflow.
Repository location: `metadata/`, `scripts/`

2. Video preprocessing and feature extraction: Videos are processed using OpenCV, with 64 frames sampled uniformly from each video. MediaPipe extracts hand, body-pose, and selected facial

landmarks. Geometric features and an availability mask are generated to represent spatial relationships and missing landmark detections.
Repository location: scripts/, models/

3. Feature encoding and model training: The custom IncludePartAwareConvHTMAV2 model uses part-aware temporal Conv1D encoders, availability-aware gated fusion, and four Head-Token Mixing Attention (HTMA) blocks to classify signs into 262 classes.
Repository location: scripts/, models/

4. Evaluation, plots, and error analysis: The trained model is evaluated on the locked test set using Top-1 accuracy, Top-5 accuracy, macro-F1 score, and balanced accuracy. The reported Top-1 accuracy is 98.04%.
Repository location: results/, scripts/

5. Demonstration and prediction: A FastAPI backend processes video predictions, while the React/Vite frontend displays the predicted sign, confidence score, and Top-5 results. Speech output is provided through the browser Web Speech API.
Repository location: app/, webcam/

6.5 Machine-Learning Implementation

The project "AI-Based Hand Gesture Recognition for Sign Language Communication" uses a custom deep learning model to classify isolated Indian Sign Language (ISL) videos into 262 classes. Since the input consists of temporal landmark and geometric features extracted from videos, conventional text-classification algorithms such as Multinomial Naive Bayes and TF-IDF-based Logistic Regression are not suitable as the primary approach.

1. Baseline and Proposed Model

Baseline model: Conv1D-HTMA baseline model.

Proposed model: Part-aware Conv1D-HTMA model with availability-aware gated fusion.

The repository contains the following training scripts:

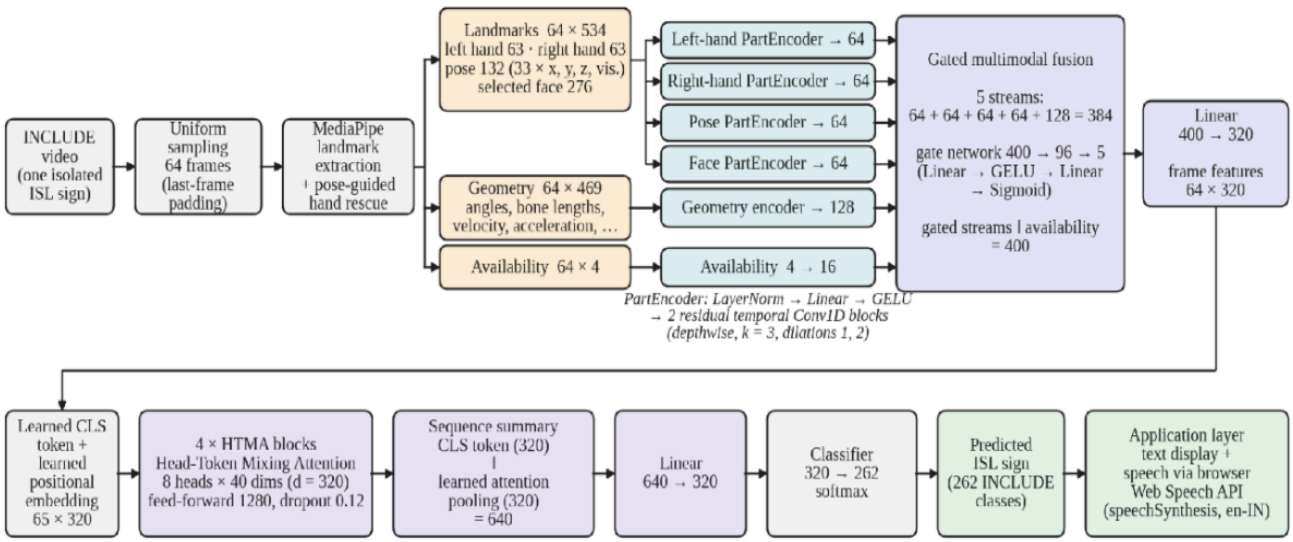
- 80_train_include262_conv1d_htma_baseline_v1_fixed.py — implements the baseline model.
- 82_train_include262_partaware_htma_v2.py — implements the part-aware model.

2. Model Architecture and Parameters

The proposed model processes landmark sequences extracted from 64 sampled video frames. Its main components are:

1. Part-aware temporal Conv1D encoders: Learn temporal patterns separately from hand, body-pose, face, and geometry features.
2. Availability-aware gated fusion: Combines the extracted feature representations while accounting for the availability of detected landmarks.
3. Head-Token Mixing Attention (HTMA): Four attention blocks model temporal relationships between frames.
4. Classification head: Produces 262 class scores corresponding to the recognized signs.

The model is trained using PyTorch with the following configuration.



Parameter	Value
Optimizer	AdamW
Initial learning rate	0.00025
Weight decay	0.0002
Batch size	32
Maximum epochs	70
Learning-rate schedule	5 warm-up epochs followed by cosine decay
Minimum learning rate	0.000001
Gradient clipping	1.0
EMA decay	0.995
Loss function	Weighted cross-entropy with label smoothing

Label smoothing	0.10
Random seed	42

The best checkpoint is selected using validation performance, and the final model is evaluated separately on the locked test set.

3. Key Code Excerpts and Module References

The following Python scripts from the project repository implement the main stages of the machine-learning pipeline.

Feature Extraction — 76_extract_include262_mediapipe64_v1.py

Extracts MediaPipe landmarks from 64-frame video sequences to represent hand movements, body posture, and selected facial landmarks.

Normalization — 78_compute_include262_train_normalization_v1.py

Computes normalization parameters from the training data to prepare features for model training.

Feature Preparation — 79_build_include262_model_features64_v1.py

Prepares the extracted landmark and geometric features in the format required by the deep learning model.

Baseline Training — 80_train_include262_conv1d_htma_baseline_v1_fixed.py

Trains the baseline Conv1D-HTMA model to learn temporal patterns from the input feature sequences.

Baseline Validation Analysis — 81_analyze_include262_baseline_v1_validation.py

Analyzes the baseline model's validation performance to assess its classification capability.

Proposed Model Training — 82_train_include262_partaware_htma_v2.py

Trains the proposed part-aware HTMA model, which uses separate feature encoders, availability-aware gated fusion, and attention blocks to learn temporal relationships between frames.

Checkpoint Verification — 83_verify_include262_v2_best_validation.py

Verifies the selected model checkpoint based on validation results.

Final Evaluation — 84_evaluate_include262_v2_FINAL_TEST.py

Evaluates the selected model on the held-out test dataset using classification metrics.

Repository: KLH-CSE-2026-27-2420030168-SignGestureAI

4. Evaluation Results

The proposed model achieved the following results on the held-out test set of 816 videos.

Metric	Result
Top-1 accuracy	98.04%
Top-5 accuracy	99.51%
Macro-F1 score	98.09%
Balanced accuracy	98.19%

These results demonstrate strong classification performance on the selected 262-class dataset. The reported metrics measure isolated-sign video classification and do not represent continuous sentence translation or live-webcam accuracy.

6.6 Demonstration / API

The project includes a web-based demonstration interface using FastAPI for the backend and React with Vite for the frontend. It allows users to submit sign-language videos and view the model's predictions.

1. Backend API

The FastAPI backend exposes a /predict endpoint that accepts a video file, processes the input, and passes the extracted features to the trained deep learning model. The model predicts one of the 262 Indian Sign Language classes.

The backend workflow includes video processing, landmark feature extraction, model inference, and returning the prediction results to the frontend.

2. Input Validation and Model Loading

The application accepts video input through the web interface. The video is processed into the required 64-frame representation, and MediaPipe extracts the necessary landmarks and geometric features.

The trained model checkpoint is loaded for inference. The input must be processed into the feature format expected by the model before prediction. The precise file-size limits, supported video formats, and validation error responses should be documented according to the actual API implementation.

3. Frontend and Prediction Display

The React/Vite frontend provides options for video upload and camera recording. After processing, it displays:

- The predicted sign label.
- The model's confidence score.
- The Top-5 predicted classes.

The browser's Web Speech API (speechSynthesis), with an en-IN voice preference, can speak the

predicted label to provide an additional accessible output.

4. Relevant Project Files

The repository's `app/` folder contains the application-related code, while `webcam/` contains webcam-related components. The `models/` folder stores model-related components, and the trained checkpoint is used for prediction.

Repository: KLH-CSE-2026-27-2420030168-SignGestureAI

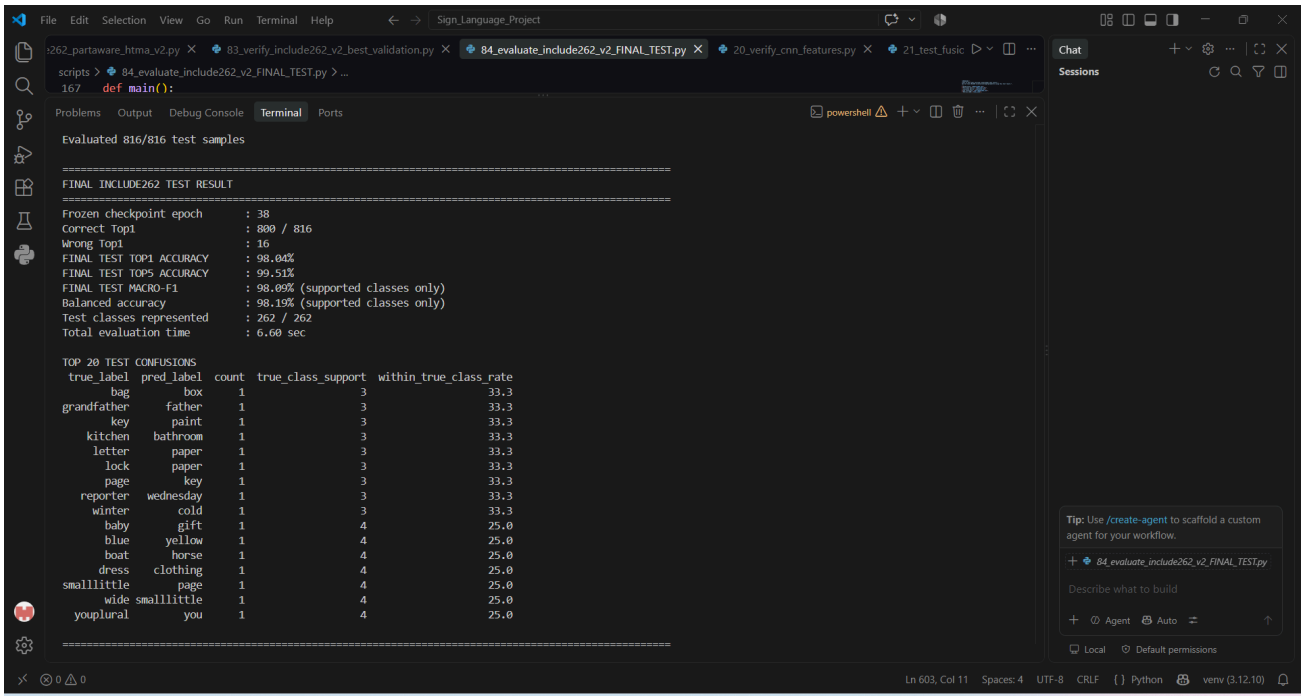
Note: The exact API source filename and checkpoint-loading function have not been verified from the source code, so they are not specified here.

5. Demonstration Workflow

1. The user uploads a sign-language video or records one using the camera.
2. The frontend sends the video to the FastAPI `/predict` endpoint.
3. The backend extracts the required features and runs inference using the trained model.
4. The API returns the predicted label, confidence score, and Top-5 results.
5. The frontend displays the results and can speak the predicted label.

The current demonstration supports isolated sign recognition. Continuous signing, sentence-level translation, and verified live-webcam accuracy remain outside the reported evaluation results.

7. RESULTS AND DISCUSSION

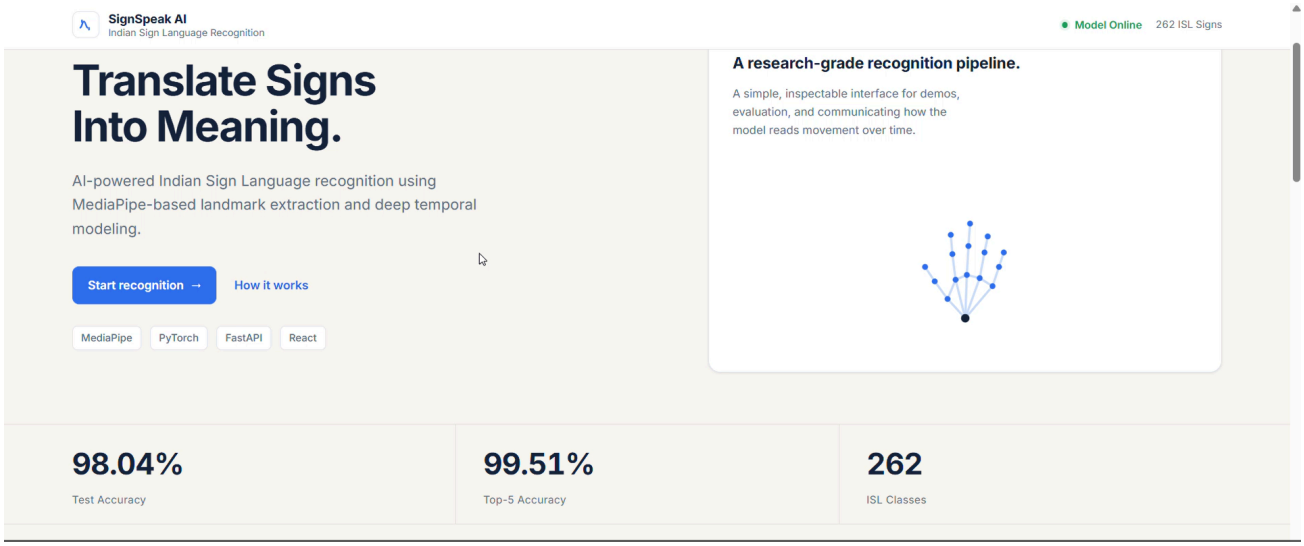


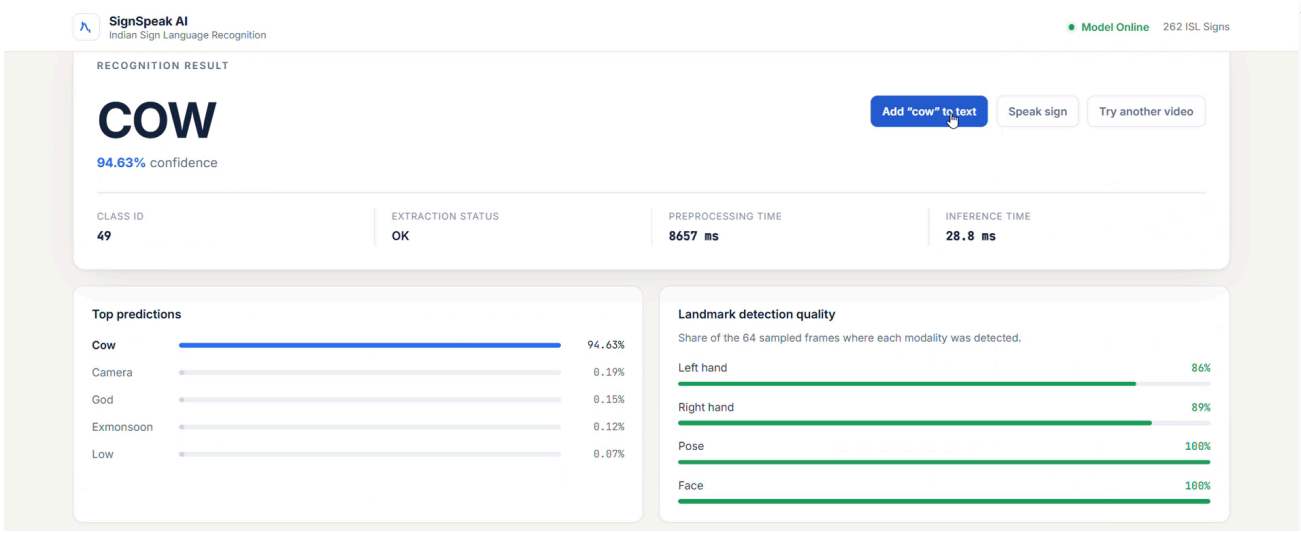
```
Evaluated 816/816 test samples

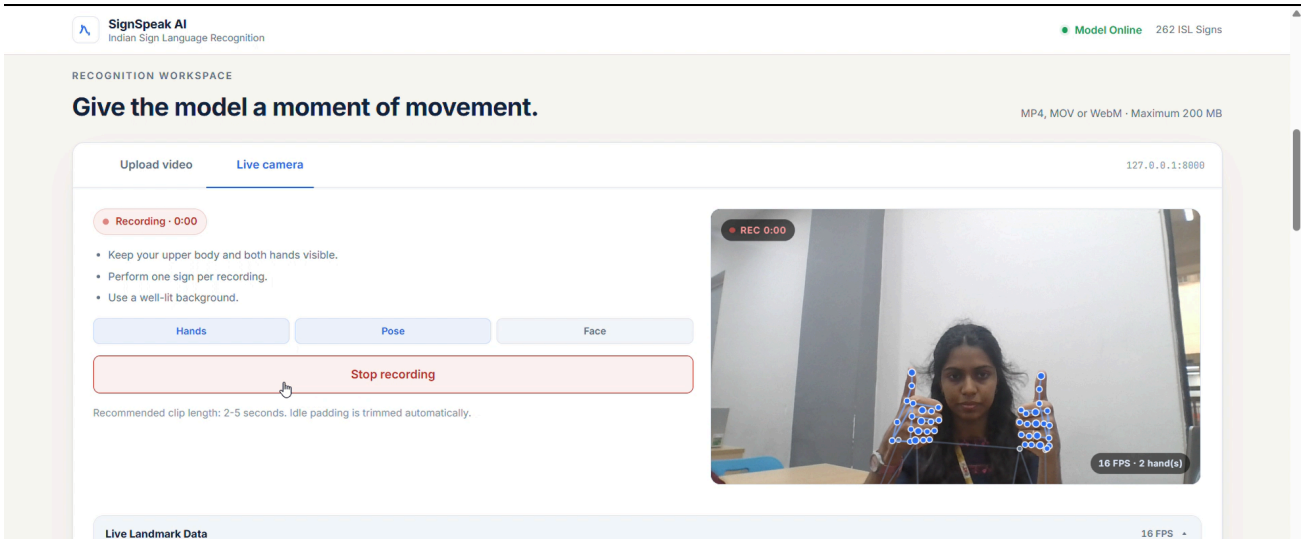
================================================================================
FINAL INCLUDE262 TEST RESULT
================================================================================
Frozen checkpoint epoch       : 38
Correct Top1                  : 800 / 816
Wrong Top1                    : 16
FINAL TEST TOP1 ACCURACY      : 98.04%
FINAL TEST TOP5 ACCURACY      : 99.51%
FINAL TEST MACRO-F1           : 98.09% (supported classes only)
Balanced accuracy             : 98.19% (supported classes only)
Test classes represented      : 262 / 262
Total evaluation time         : 6.60 sec

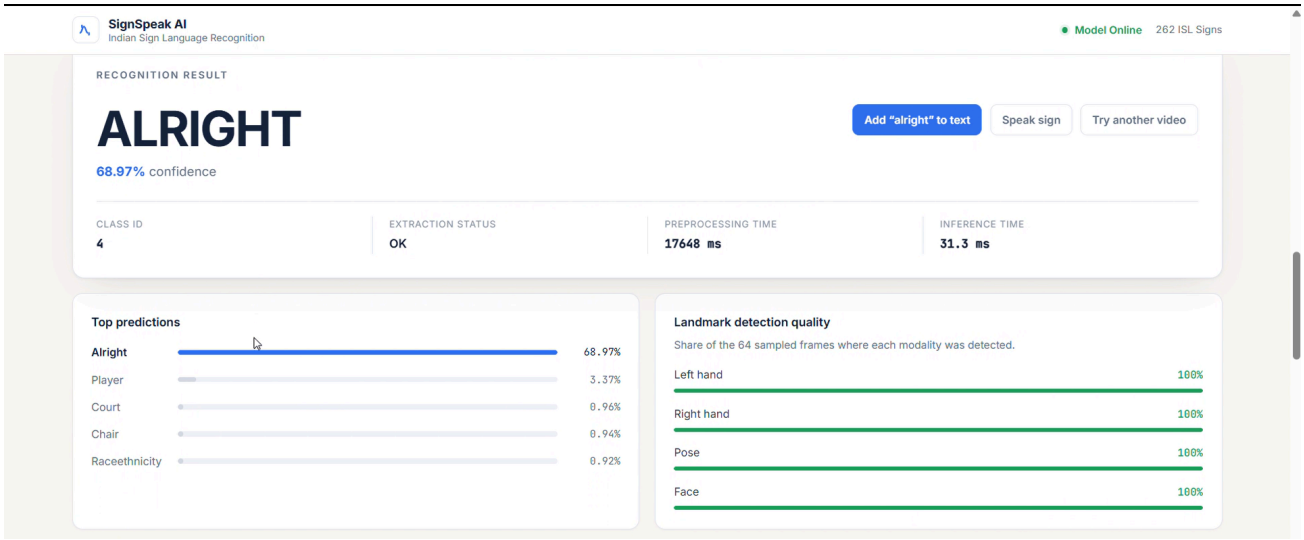
TOP 20 TEST CONFUSIONS
 true_label  pred_label  count  true_class_support  within_true_class_rate
        bag         box      1                   3                    33.3
 grandfather      father      1                   3                    33.3
        key       paint      1                   3                    33.3
    kitchen    bathroom      1                   3                    33.3
     letter       paper      1                   3                    33.3
       lock       paper      1                   3                    33.3
       page         key      1                   3                    33.3
   reporter   wednesday      1                   3                    33.3
     winter        cold      1                   3                    33.3
       baby        gift      1                   4                    25.0
       blue      yellow      1                   4                    25.0
       boat       horse      1                   4                    25.0
      dress    clothing      1                   4                    25.0
 smalllittle        page      1                   4                    25.0
       wide smalllittle      1                   4                    25.0
  youplural         you      1                   4                    25.0

================================================================================
```




SignSpeak AI
Indian Sign Language Recognition
Model Online
262 ISL Signs
RECOGNITION RESULT
COW
94.63% confidence
Add "cow" to text
Speak sign
Try another video
CLASS ID
49
EXTRACTION STATUS
OK
PREPROCESSING TIME
8657 ms
INFERENCE TIME
28.8 ms
Top predictions
Cow 94.63%
Camera 0.19%
God 0.15%
Exmonsoon 0.12%
Low 0.07%
Landmark detection quality
Share of the 64 sampled frames where each modality was detected.
Left hand 86%
Right hand 89%
Pose 100%
Face 100%


SignSpeak AI
Indian Sign Language Recognition
Model Online
262 ISL Signs
RECOGNITION WORKSPACE
Give the model a moment of movement.
MP4, MOV or WebM · Maximum 200 MB
Upload video
Live camera
127.0.0.1:8000
Recording · 0:00
Keep your upper body and both hands visible.
Perform one sign per recording.
Use a well-lit background.
Hands
Pose
Face
Stop recording
Recommended clip length: 2-5 seconds. Idle padding is trimmed automatically.
REC 0:00
16 FPS · 2 hand(s)
Live Landmark Data
16 FPS


SignSpeak AI
Indian Sign Language Recognition
Model Online
262 ISL Signs
RECOGNITION RESULT
ALRIGHT
68.97% confidence
Add "alright" to text
Speak sign
Try another video
CLASS ID
4
EXTRACTION STATUS
OK
PREPROCESSING TIME
17648 ms
INFERENCE TIME
31.3 ms
Top predictions
Alright 68.97%
Player 3.37%
Court 0.96%
Chair 0.94%
Raceethnicity 0.92%
Landmark detection quality
Share of the 64 sampled frames where each modality was detected.
Left hand 100%
Right hand 100%
Pose 100%
Face 100%

UNDER THE HOOD

How SignSpeak AI Works

A compact multi-stream representation gives the model enough context to read movement, geometry and landmark availability over time.

64 frames sampled	534-D landmark representation
469-D geometry representation	4-D availability mask
262 sign classes	

98.04%

Final test accuracy

01 **Video input**

A short MP4 or MOV clip is uniformly sampled into 64 temporal frames.

02 **Landmark extraction**

MediaPipe tracks hands, pose and selected facial landmarks into a 534-D landmark representation.

03 **Geometry features**

Motion and spatial relationships create a 469-D geometry representation.

04 **Availability mask**

A 4-D availability mask tells the model which modalities were visible in each frame.

05 **Part-aware temporal encoding**

Part-specific Conv1D encoders process left hand, right hand, pose, face and geometry features.

06 **HTMA temporal recognition**

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project “AI-Based Hand Gesture Recognition for Sign Language Communication” was developed to support communication by recognizing isolated Indian Sign Language (ISL) signs from video and presenting the predicted labels as text and speech.

The system uses the INCLUDE dataset, with 262 sign classes and official training, validation, and test partitions. OpenCV is used to sample 64 frames from each video, while MediaPipe extracts hand, body-pose, and selected facial landmarks. Geometric features and an availability mask are combined with a custom deep learning architecture consisting of part-aware temporal Conv1D encoders, availability-aware gated fusion, and Head-Token Mixing Attention (HTMA) blocks.

The implementation includes a baseline Conv1D-HTMA model and the proposed part-aware HTMA model. The final model was selected using validation performance and evaluated on a separate test set of 816 videos.

The final model achieved 98.04% Top-1 accuracy, 99.51% Top-5 accuracy, 98.09% macro-F1 score, and 98.19% balanced accuracy, with 800 correct predictions out of 816 test videos. These findings demonstrate strong performance on the selected isolated-sign classification task.

The project objectives were achieved in the areas of dataset preparation, landmark feature extraction, temporal model development, test-set evaluation, and web application integration. The FastAPI backend and React/Vite frontend provide video upload and camera-recording options, prediction display, and optional speech output.

The most important limitation is that the system currently recognizes isolated signs rather than continuous signing or complete sentences. In addition, the reported test-set metrics do not establish real-world performance across different users, recording conditions, or live-webcam inputs.

8.2 Future Scope

The project can be extended in the following ways:

1. Continuous Sign Language Recognition: Extend the system to recognize sequences of connected signs and translate them into meaningful sentences.

2. Larger and More Diverse Datasets: Evaluate and train the model using additional signers, regional variations, backgrounds, lighting conditions, and recording environments to improve generalization.

3. Class Imbalance and Error Analysis: Investigate frequently confused sign classes and improve performance on underrepresented classes through suitable sampling and training strategies.

4. Real-Time Webcam Evaluation: Measure prediction latency, processing speed, and accuracy during live camera use to assess practical deployment readiness.

5. Model Optimization: Explore efficient temporal architectures and model compression to reduce inference time and computational requirements.

6. Explainability: Develop methods to identify the frames, landmark regions, or movement patterns that contribute most to a prediction.

7. Privacy-Aware Processing: Investigate local video processing and secure handling of uploaded videos to reduce unnecessary exposure of sensitive visual data.

8. Deployment and Usability Testing: Deploy the application in a suitable environment and evaluate its usability, reliability, and accessibility with users under realistic conditions.

These extensions could help improve the system's robustness, usability, and suitability for practical Indian Sign Language communication support.

9. REFERENCES

The following references are relevant to the project "AI-Based Hand Gesture Recognition for Sign Language Communication." They cover the INCLUDE dataset, sign-language recognition research, landmark-based and Transformer-based methods, and the software frameworks used in the implementation.

Since this project performs video-based sign classification rather than text analytics, references to NLTK, spaCy, TF-IDF, and BERT are not required unless they were actually used.

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